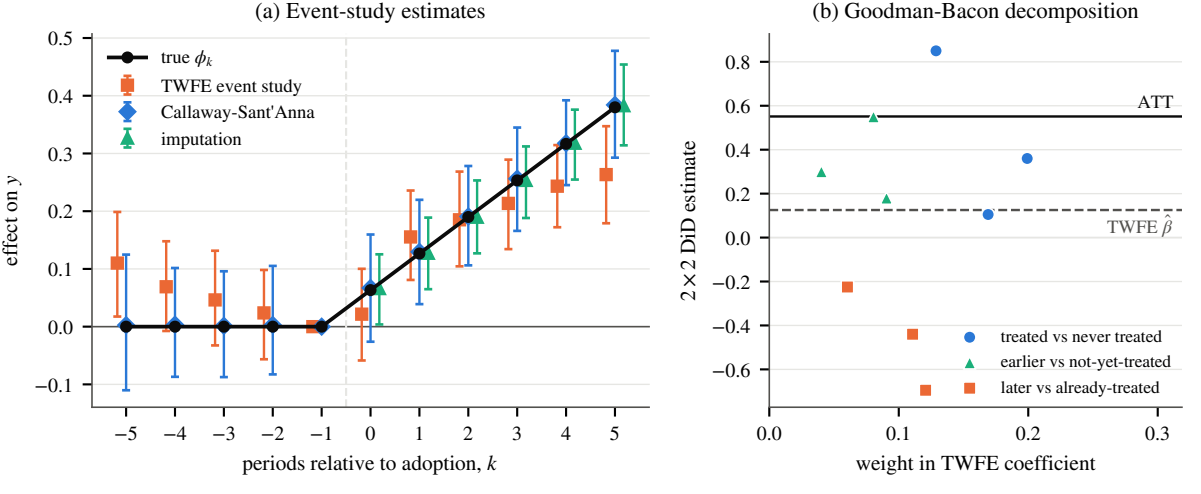
(a) Event-study estimates
true ϕ_k
TWFE event study
Callaway-Sant'Anna
imputation
effect on y
periods relative to adoption, k
(b) Goodman-Bacon decomposition
ATT
TWFE $\hat{\beta}$
treated vs never treated
earlier vs not-yet-treated
later vs already-treated
2×2 DiD estimate
weight in TWFE coefficient